Better. Brighter.

USER REQUIREMENT SPECIFICATIONS

ICOMS 2.0

Module 3 : TOMS

Contents

1.0 DOCUMENT PURPOSE

The purpose of this document is to define the **User Requirements Specification (URS)** for the GNM functionalities within the ICOMS 2.0 system.

This document describes the system behaviour, workflows, and functional requirements related to the **monitoring, study, coordination, and approval** of outages by GNM users. It outlines the user interfaces, operational processes, and system capabilities required to support outage assessment, approval management, repository access, reporting, and system-wide coordination activities.

The document serves as a reference for system development, configuration, and validation to ensure that the ICOMS 2.0 system supports the operational needs of GNM users in managing and analysing outage activities across the network.

2.0 OBJECTIVES

The objective of this document is to define the functional requirements that enable GNM users to monitor, review, analyse, and manage outages within the ICOMS 2.0 system.

The key objectives include:

1. To define the system functionality required for reviewing, studying, and approving outages by GNM users.
2. To establish the outage workflow and status progression under TOMS control, including Pending, Under-Study, KIV, Approved, and Disapproved statuses.
3. To describe the processes for outage assessment, including technical study inputs such as justification, highlights, remarks, and off-point suggestions.
4. To define the functionality for managing outage repositories with advanced filtering, sorting, and bulk approval capabilities.
5. To provide capabilities for identifying conflicting outages, analysing equipment history, and accessing supporting system data during outage evaluation.
6. To support coordination activities including GCU notification, outage scheduling visibility, and interdependency management.
7. To define the reporting and statistical capabilities required for outage analysis, performance tracking, and operational planning.
8. To provide functionality for managing Single Line Diagram (SLD) creation, updates, and approvals within the system.
9. To provide functionality for managing Comm Memo creation, updates, and approvals within the system.
10. To support system-wide monitoring of outage authorisation status and repository access for ongoing and completed outages.
11. Library to store and view data forms for substations – downloadable PDF listings of Dataforms

3.0 SCOPE

This document covers the TOMS user functionalities within the ICOMS 2.0 system.

The scope includes the following functional areas:

1. **Outage Repository and Monitoring**
   Viewing and managing a comprehensive repository of all outages with advanced filtering, sorting, and preset configurations, including access to detailed outage information and system-wide visibility.
2. **Outage Study and Evaluation**
   Reviewing outages under TOMS responsibility, including input of study data such as justification, highlights, remarks, under-study notes, and off-point suggestions.
3. **Outage Approval Management**
   Management of outage approval workflows including reviewing Pending outages, conducting studies, and assigning statuses such as Approved, Disapproved, or KIV, with support for bulk approval actions.
4. **Outage Coordination and Conflict Management**
   Identification and handling of conflicting outages, coordination with GCU users, and management of interdependent outage activities across the network.
5. **Outage Status and Workflow Control**
   Control and modification of outage statuses within the TOMS scope, including the ability to transition outages between study stages prior to authorisation.
6. **Reporting and Customised Analysis**
   Generation of customised reports using system filters, export functionality, and the ability to save user-defined report configurations.
7. **Statistics and Performance Analysis**
   Access to predefined and custom statistical views, including historical, current, and forecast outage data, as well as analytical tools for identifying trends and repetitive outages.
8. **Authorisation Repository Monitoring**
   Viewing and tracking outages with Taken-Active and Taken-Completed statuses, including access to authorisation history and operational timelines.
9. **Outage Request Creation (On Behalf of Requestor)**
   Creation of outages by TOMS users on behalf of Requestors, including automatic workflow progression and coordination with Requestor confirmation.
10. **Single Line Diagram (SLD) Management**
    Creation, submission, review, approval, and system-wide publication of new and updated Single Line Diagrams, including integration with outage-related processes.
11. **System Configuration (Database Maintenance Reference)**
    Access to system-wide configuration settings managed by GNM Admin users, as referenced in Module 1.
12. **Comm Memo Management**
    Creation, submission, review, approval of the Commissioning memo.

4.0 LIST OF USERS ROLES

No	Name	Division
1	GNM ADMIN	
2	GNM	

3		
4		
5		
6		

5.0 USER REQUIREMENT

5.1 Menu Update

Legacy	ICOMS 2.0	Remark
ADMIN MODULE		
Maintenance Outage Review	Removed	Removed - This will be Moved to Outage Repository. Will show all outages, user will be able to filter by Outage Type
Project Outage Review		
Generator Outage Module		
Database Maintenance	Database Maintenance	Will have all setup for the system - subject to TOMS ADMIN role.
OUTAGE STATUS		
Outage Repo	Outage Repository	Combine into one, outage status, showing all outages. User will have dropdowns to select different job types, status, planned/unplanned/emergency/forced. There will need to be a checkbox next to all listings, allowing user to select all and approve all.
Forced Outage List		
System Status	Removed	Removed
	Outage (Docket) Approval	Shows the next day, next week outages as pre-set, along with checkbox for mass approval and output to report as Excel, Pdf and other valid formats. There will be 5 tabs showing • Tomorrow • Saturday-Sunday-Monday upcoming • Next Week • Next Week Friday-Saturday-Sunday • Next Month (all 500kV) + Month After (all 275kV and 500 kV)
REPORTING MODULE		
Coordination Report Page		Removed - Shifted as Outage Approval in the Outage Status Module.
Customised Report Page	Customised Report Page	Full search and export functionality, allow users to select all available filters to customer create their own reports.
Statistics	Statistics	Full Stats as per requested docs
Calendar	Calander	
AUTHORISATION MODULE		
Authorisation in Force	Authorisation	Shows all Authorisation that are Taken-Active

Long Authorisation	Repository	Shows all Outages that are over 7 days, needs to have one Taken-Active authorisation
Authorisation Repository		All Authorisations
REQUEST MODULE		
Planned Outage	Request Outage	Will be merged into one request page. The system will auto set the Planned/Unplanned/Emergency tag based on the outage period selected.
Unplanned Outage		
Emergency Outage		
NEW MODULES		
	Single Line Diagram	
	Commissioning	
SPECIAL PAGE		
PMU Installation Management		Removed

5.2 Database Maintenance

This is system wide setup by GNM Admin users, the details of which have been captured in URS Module 1 document.

5.3 Outage Repository

TOMS will have a full repository of outages with complete filtering to show job types (Project, Maintenance, Generator etc.); Voltage and Equipment; Agreed and Confirmed; all status selection; Outage Type (Planned, Unplanned, Emergency, Forced); among others. TOMS users will be able to sort through these filters and generate the exact listing they need at the time. There will be a checkbox selection, with the ability to select all and Approve All at the same time (only in the outage docket). By default, the system will show all outages, even those that are Unconfirmed or have not been Agreed to by the Planners. This will be shown for the current month and year, though the user can edit the selection to show previous and upcoming outages.

Each user will have the option to save a preset of these filters for their account, which they can click at any time to generate the listing. These saved preset filters will be tied to individual accounts, and will be managed by the users. This will allow TOMS the flexibility to store their most used listing and be able to display them with one click.

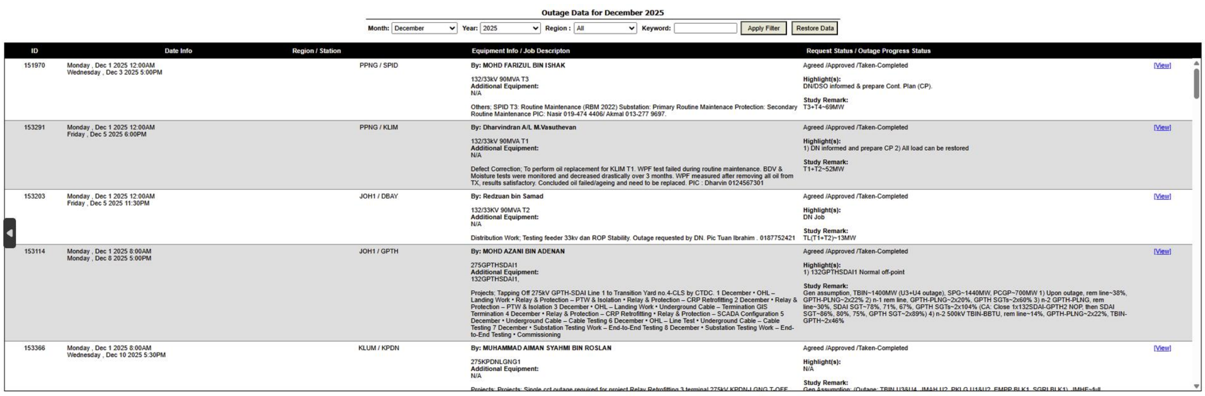


Figure 5.3a Outage Listing

Enhancements to the outage listing

- The outage listing will show the Restoration time
- The outage listing will also show the list of PIC (Names only), underneath the description


REQUEST ID: [151970] Update | Update & Notify | Cancel

EQUIPMENT INFORMATION:
Region: Pulau Pinang
Station: Sungai Petani Industrial
Voltage: 132
Filter(Line): Not Applicable
Equipment: Transformer
132/33kV 90MVA T3
Additional: No Additional Selected

HISTORICAL INFORMATION (LAST EDITED):
Date: Outage request hasn't been edited
By:

CONTROLLER'S STATUS:
Status: Agreed
Comment: No Comment Provided
By: MOHD FARIZUL BIN ISHAK

OUTAGE DATE:
Start: 01-Dec-2025 12:00 AM
End: 03-Dec-2025 5:00 PM
Code: Unplanned [Check Conflicting Outages]

ADDITIONAL INFORMATION:
Restoration: 30 Mins
Sequence: Not Applicable.
Job Type: Others
Work Type: Dead
Daily PTW: Daily PTW
Weightage: 1 [Reassign Weightage]
Description: SPID T3: Routine Maintenance (RBM 2022)
Substation: Primary Routine Maintenace
Protection: Secondary Routine Maintenance
PIC: Nasir 019-474 4406/ Akmal 013-277 9697.

APPROVAL SECTION:
Status: Approve Denied in Docket Outage Confirmed?
Justification: Outage date change from 12-14 Nov 25 to 1 - 3 Dec 25. Wassap from Ikram on 27/11/25 @2145hrs.
Highlight(s): DN/DSO informed & prepare Cont. Plan (CP).
Remark: T3+T4~69MW
[Check Study Remark History]
Date: Tuesday, December 30, 2025
By: Anas bin Sheikh Zaki

REQUESTER INFORMATION:
Name: MOHD FARIZUL BIN ISHAK
Organisation: AM-PPNG

OUTAGE SLOT INFORMATION:
[Refresh Date] [View Detailed Slot]
Slot for Monday, December 01, 2025 : 15

CHECK RESULTS

CHECK SLD:

Voltage Level	SLD
132	SPID.pdf

Update | Update & Notify | Cancel

Figure 5.3b View and Edit Outage

Enhancement – Additional data to be shown on the detail view stage for every outage

- Link to SLD of the Station and Line that is selected – In the information sidebar; pdf will be opened in a new tab, available for download
- List of other outages with Line equipment that are conflicting with the current Line outage. It will show all outages that are approved/active before the date of the outage being studied, which will potentially interfere with the current Outage. In case of outages being extended by the GNC or not being closed and pushed to red status, the system will check the lines of that outage to see if they have conflicting outages on those lines coming up in the extended timeline. If there are outages that are conflicting, the TOMS team will be able to deal with it appropriately, either by adjusting the time or shifting it to KIV for later.

- There should be suggest Off-point dropdown. This is done by clicking the Off-Point link which will open a pop-up allowing users to select the Zone, Station and pick from a list of existing Off-points or suggest a temporary one with the existing equipment. There will text box for off points remarks for GNC, which they can use to suggest Open/close,
- Users should be able to see the history of the outages the equipment has gone through. By clicking the link, it will show all the outage history along with the job type of the outage. Clinging the outage ID will open the detailed view of the previous outage. This will be in the information sidebar. Last Maintenance will show the data from the last 10 years of only the Job Type Maintenance for the selected equipment. The Outage history for all other job types will show all other job types with Outage Closed status for the last 3 years.

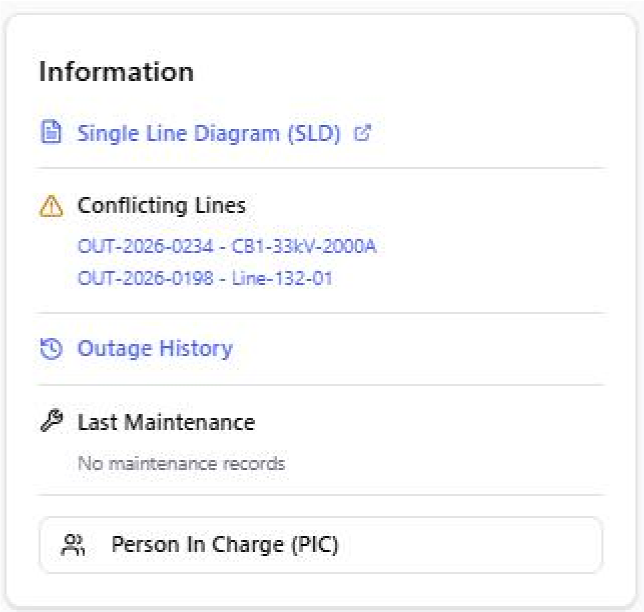


Figure 5.3c Information Sidebar

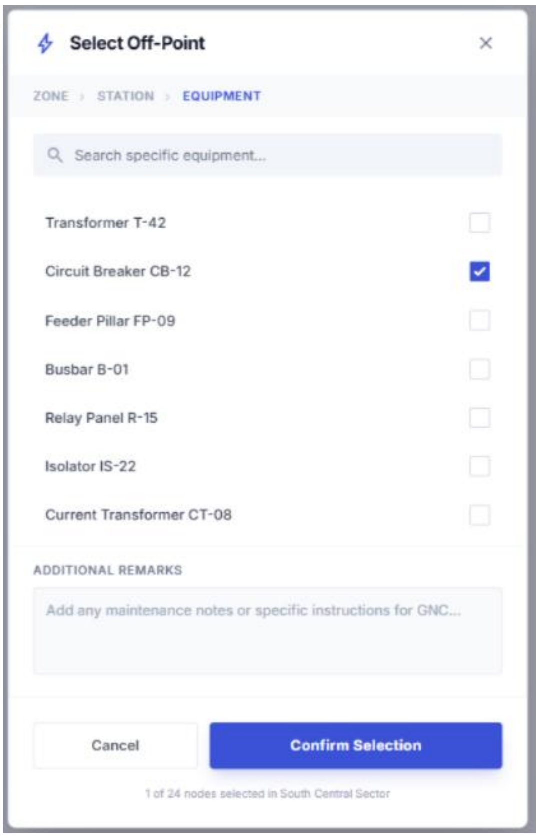


Figure 5.3d Off-Point Selector

TOMS will use the view/edit functionality to change the Status of the Outages, along with adding all their study findings in the appropriate fields.

Field	Field Type	Remarks
Justification	Text Box	TOMS to fill as needed
High-Lights	Text Box	TOMS to fill as needed
Remark	Text Box	TOMS to fill as needed
Under-study	Text Box	TOMS to fill as needed
Off-Point Suggestion	Pop-up Link	Opens pop-up to select Off-points
Date Edited	Auto-filled	Date of Edit
Edited by	Auto-filled	TOMS user
Status	Dropdown	Pending, Understudy, Approved

For an outage that has been Taken-Active status, TOMS can only edit the description and the Study Section under the headings Justification, Under-study, Highlight and Remarks.

Users will have the option to Update the listing or Update and Notify, which will send an email to the PIC notifying them about edits made. There will be a conformation for the Update and Notify option, allowing the users to add more emails to the list. These additional emails will be saved so the next time the Update and Notify button is pressed, there are there by default. The complete edits will not be shown on the email; the PIC users will have to login to the system to view the details.

If TOMS sets an outage status to KIV, it is taken out of the pool of active outages that may cause conflicts with new/pending outages. The KIV will be a separate list of outages that are to be edited later. Requestors will be able to see the KIV status on their outages and submit change requests. If an outage is kept in the KIV status for one month, it will automatically be given the status OUTAGE CLOSED – KIV TIMEOUT and be removed from the listing.

In case TOMS does not shift the status of a pending/understudy outage to APPROVED within the first 24 hours of the outage start date and time, it will shift to the KIV status.

5.4 Outage Approval

When an Outage has been Agreed by the Planner and Confirmed by the Requestor, its status shifts to Pending and is now under the work scope of TOMS team. TOMS will study the outage, shift the status to Under-Study, add notes, and eventually must either Approve or Disapprove the outage.

The Outage Approval page shows five dockets for the TOMS list to push the Approval or Disapproval through.

- Next Day
- Saturday-Sunday-Monday upcoming
- Next Week
- Next Week Friday-Saturday-Sunday
- Next Month (all 500kV) + Month After (all 275kV and 500 kV)

Each tab will show its representing outages for mass approval by the multiselect checkbox system. The tabs will have the filters of the Outage Repository allowing the users to filter different zones/types and statuses. Users will be able to print this docket to a pfd file as necessary.

TOMS can backtrack the status from Approved. They can shift the status back to Under-Study or Pending and then back to Approved if the outage does not have a Taken-Active status set by GNC.

To facilitate quick Approval, there will be a check box functionality with Select All, Approve All and Reject All options.

Check All | Uncheck All | Agree Checked Items | Disagree Checked Items

Confirmed?	ID	Start / End Date	Station	Equipment Info	Description	Action
	155203	Monday, Apr 13 2026 8:30AM Monday , Apr 13 2026 5:00PM **Code : U**	BTRZ (AM-KLUM)	**Voltage Level:**132 132BTRZPULU1 **Additional Equipment:** N/A	**Defect Correction** DEFECT RECTIFICATION 1. HOTSPOT P1 AT TOP PHASE 132KV BTRZ-SPKC TOFF MCRS T31/25. (38 'C)	[Review]
	155204	Wednesday, Apr 15 2026 12:00AM Wednesday , Apr 15 2026 12:00AM **Code : U**	NUNI (AM-KLUM)	**Voltage Level:**275 132ABBANUNI1 **Additional Equipment:** N/A	**Defect Correction** HOSPOT RECTIFICATION T9, BOTTOM PHASE P3 PIC SYAFIQ - 0166671990	[Review]
	155207	Monday, Apr 13 2026 8:30AM Monday , Apr 13 2026 6:00PM **Code : U**	BTRZ (AM-KLUM)	**Voltage Level:**132 132BTRZ-SPKC/MCRS2 **Additional Equipment:** N/A	**Defect Correction** DEFECT CORRECTION HOTSPOT AT T31/25. P1 HOTSPOT (38.7 'C) PIC - SYAFIQ 0166671990	[Review]
	155210	Tuesday, Jul 21 2026 8:30AM Tuesday , Jul 21 2026 6:00PM **Code : U**	SRDE (AM-KLUM)	**Voltage Level:**275 275kVSRDELGNG1 **Additional Equipment:** 132NUNIIOIM2,	**Defect Correction** OUTAGE QUAD TOWER 275KV SRDE-LGNG L1 / 132KV NUNI-IOIM L2 HOTSPOT (P3) PIC SYAFIQ - 0166671990	[Review]
	155227	Tuesday, Apr 7 2026 9:00AM Tuesday , Apr 7 2026 10:00PM **Code : U**	KCPK (AM-KLUM)	**Voltage Level:**132 132/33kV 90MVA T1 **Additional Equipment:** N/A	**Others** Surge Arrestor Replacement Works (Initiatives SUBD GMKL 2026). Jobsheet DOMS SEL-26-03105. PIC GMKL Hilmi (011-10320261)	[Review]
	155228	Tuesday, Apr 14 2026 9:00AM Tuesday , Apr 14 2026 10:00PM **Code : U**	KCPK (AM-KLUM)	**Voltage Level:**132 132/33kV 90MVA T2 **Additional Equipment:** N/A	**Others** SA replacement Works (Initiatives SUBD GMKL 2026) Jobsheet DOMS SEL-26-03108 PIC GMKL Hilmi (011-10320261)	[Review]
	155243	Monday, May 11 2026 9:00AM Tuesday , May 12 2026 10:00PM **Code : U**	CBJN (AM-KLUM)	**Voltage Level:**132 132/33kV 90MVA T3	**Others** Retest HV bushing CBJN T3 (Initiatives SUBD GMKL) Jobsheet DOMS PTJ-26-00126 PIC GMKL Hilmi (011-	[Review]

Figure 5.4a Listing Screen showing outages to be Approved or Disapproved

TOMS can push the outage to a GCU user for acknowledgement, by entering the detailed view of the outage and pressing the “Push to GCU” and selecting the GCU user assigned to the zone of the outage.

5.5 Customised Report Page

This page will allow users to generate their own listing screen, by selecting any available filter options available within the system. The generated listings can then be exported out as PDF and Excel files as reports for the users.

Users will be able to save their selected filters as a Favourite. This will enable them to generate reports with one click. Users are allowed to save multiple filters as favourites, tied to the individual account.

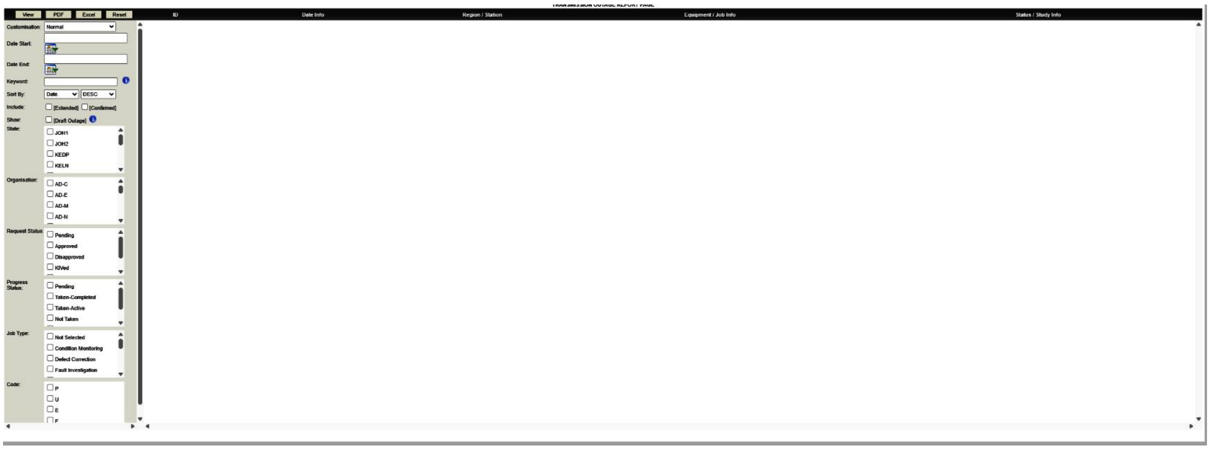


Figure 5.5a Generate listing based on filters

5.6 Statistics

The statistics menu will show list of different reports and graphs available for TOMS to review. TOMS will be able to create a new chart using the customisation option, or select one of the pre-loaded options to display the required information. These will be divided into different tabs for ease of use. The generated charts and graphs will be downable in JPEG format. All reports generated can be downloaded in either Excel or PDF format.

5.6.1 Customised

This tab will allow users to create customised set of bar charts. By selecting the correct x and y axis, the system will display the information requested. For every x-axis selected, there will be a set of corresponding y-axis available. These will be detailed out during the development process and will require the review and approval of TNB GSO management.

5.6.2 TYTOP

This tab will show the default charts as listed below.

One set of charts will show the current (y) and previous 3 years information (y-1, y-2 and y-3) for a total of 4 years.

By Year/Month	Type of Chart	Type of chart
Year, Month	Number of requested Planned Outage for each organisation - outages all (regardless whatever status)	Bar Chart
Year	Approved Outage for Planned, Unplanned, Emergency & Forced Outage & comparison for 3 years data	Bar Chart

Year	Approved Outage for Planned, Unplanned, Emergency & Forced Outage & comparison for 3 years data and according to voltage level	Bar Chart
Year	Approved Outage for Planned, Unplanned, Emergency & Forced Outage for each organisation and according to voltage level	Bar Chart
Year	Actual Outage Taken for Planned, Unplanned, Emergency & Forced Outage	Pie Chart
Year	Actual Outage Taken for Planned, Unplanned, Emergency & Forced Outage by organisation and according to voltage level	Bar Chart
Year, Month	Total of Outage Not taken	Bar Chart
Year, Month	Total of Outage Not taken by organisation	Bar Chart
N/A	Reason of Outage Not Taken	Pie Chart
Year, Month	Total Outage rescheduled for each organisation	Bar Chart
Year, Month	Total Outage rescheduled for each organisation by GNC (cancel by GNC +have Change Request form only for Planned outage)/GNM/Requester	Bar Chart
Year, Month	Total Planned outage used up for the year & percentage based on the number requested planned outage keyed in from Jan-June (Agreed + Confirmed)	Bar Chart
Year, Month	Total outage taken according to job type	Bar Chart
Month	Total Outage taken by Month (Jan -Dec in 1 chart)	Bar Chart
Month	Total Outage taken by Month (Jan -Dec in 1 chart) by each organisation	Bar Chart
Month	Total Outages taken-active/taken-completed for each organisation	Bar Chart
N/A	Total number of equipment created by voltage level (e.g.: busbars, lines, bays)	Bar Chart
Month	Total extended outages filtered by region/voltage levels in the current year	Bar Chart
N/A	Total number of outages extended based on job type	Bar Chart

Another set of charts will show the information for the following three years (y+1, y+2, y+3), along with the Total Outage Hours, and Planned Availability.

By Year/Month	Type of Chart	Type of chart
Year, Month	Number of requested Planned Outage for each organisation - outages all (regardless whatever status)	Bar Chart
Year, Month	Total Requested Planned Outage by spread out by month throughout the year	Bar Chart
Year, Month	Total Requested Planned Outage by job type	Bar Chart
Year, Month	Total Requested Planned Outage by job type and by each organization	Bar Chart

Total Outage Hours
User will select the intended year (y+1, y+2, y+3), which will display a list of

- All 132kv outages that have Lines, SGT
- All 275kv and 500kv outages that have Lines and Interbus Transformer

The system will show the outage hours of each of the individual outages and then calculate the Total Outage Hours from the list.

Planned Availability
This will be calculated using the following formula once the user has input the required fields.

- n
- Additional TOH

PA = 100 x (1-((TOH from List) + Additional TOH) / (n x 24 x 365)))

5.6.3 Analysis

There will be one horizontal bar chart, showing the repetitive outages for the current year. The y-axis will show the equipment, with the x-axis showing the number of outages. This chart will list the equipment in descending number of outages, with the highest on top. Users will be able to click on the bar to see the list of outages by job type for each equipment and its ID

There will also be a section for users to select an equipment and view its last outage date along with its first upcoming outage date.

5.7 Calander

Refer to “Module 2: Requestor and Planner” document for requirements.

5.8 Authorisation Repository

The Authorisation Repository will show a complete listing of all outages that have the status **Taken-Active** and **Taken-Completed**. The outages will leave this list when they are given any one of the **Outage Closed** status. Users will be able to “View Authorisation Workplan” to see a complete history of the outage plan by GNC and subsequent details.

The listing screen will be colour coded to show the timing of the outage. Green will be those are currently in the outage timeline window. Yellow will be for those that have been extended by GNC users. Red will denote the outages where the original time has been over, but they still have a Taken-Active status, and have not been closed with any of the Outage Closed status

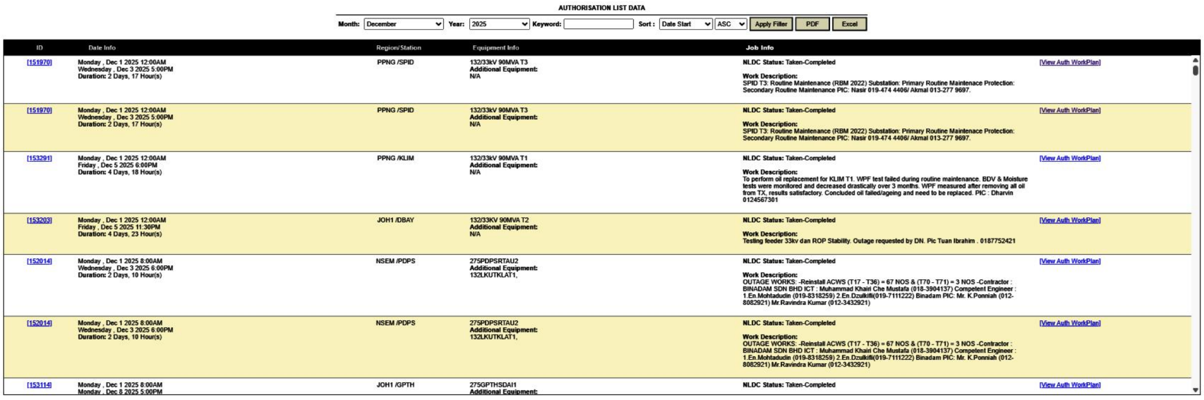


Figure 5.8a Authorisation Repository

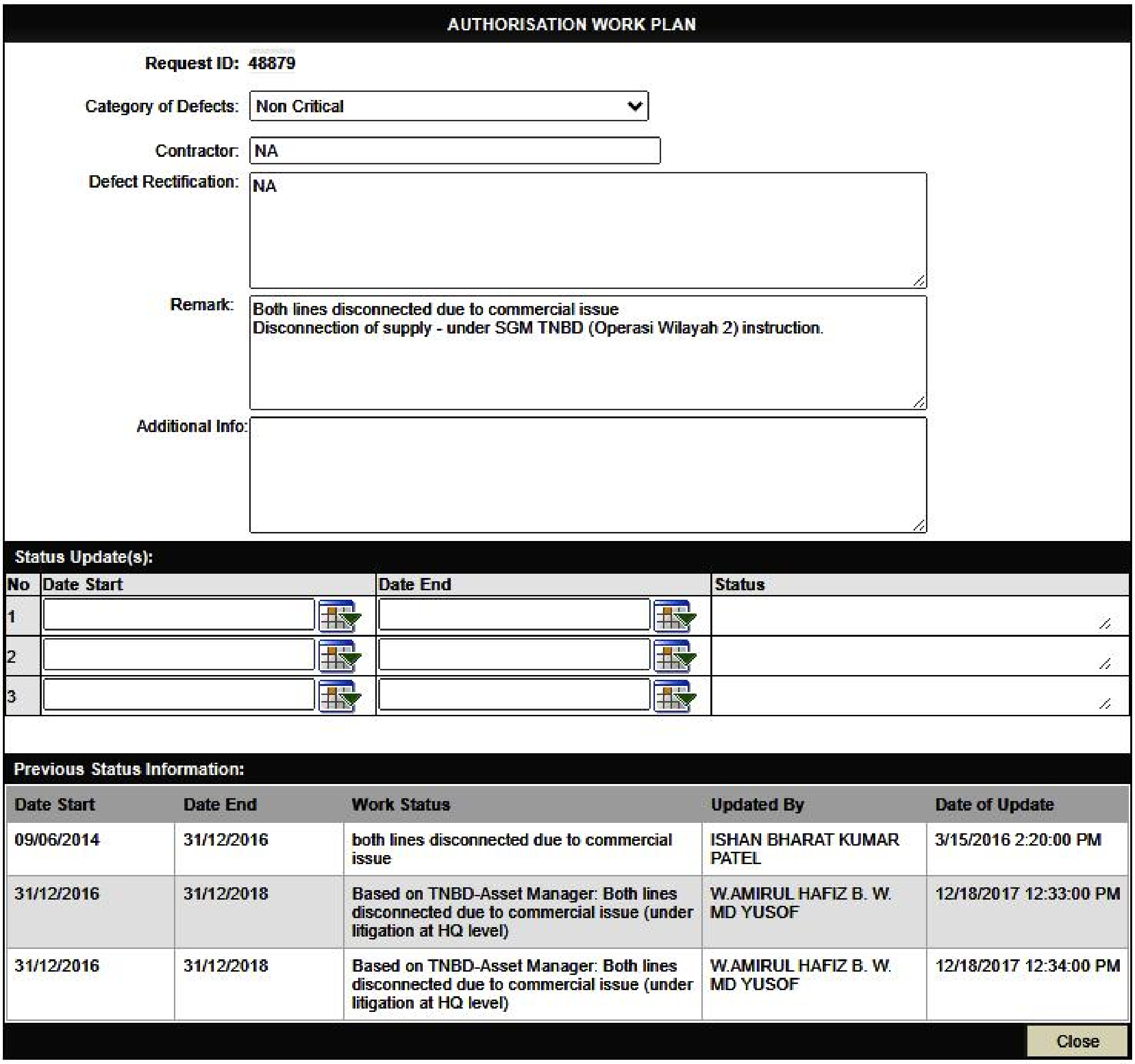


Figure 5.8b View Authorisation Workplan

5.9 Request Outage

This section will allow TOMS to create outages on behalf of Requestors. TOMS will need to input the details of the Requestor in the PIC section of outage creation. All Outages created by TOMS are automatically given the Agreed (Planner) status. The Requestor in the PIC will need to give the Confirmed status for the outage to move into Pending for TOMS.

The complete URS for Outage creation is detailed in Module 2: Requestor and Planner document.

5.10 Single Line Diagram

The Single Line Diagram is a new module in ICOMS 2.0 that will allow users to create a new SLD to be saved in the system, tied to a station. Once a station is selected in the outage creation, the SLD will be available for download from the right-hand information sidebar.

The users can also update an existing SLD or update a newly commissioned. All approval and acknowledgements within the workflow will attachments to be uploaded along with the compulsory comments for user review.

Workflow will be as follows.

5.10.1 New SLD

1. The REQUESTOR will select "New SLD" and input the following details
    a. Region
    b. KV Level (For every PMU selected)
    c. New PMU Name + New SLD
    d. Existing Remote End PMU/s + Updated SLD for the existing remote end PMU
    e. Attach Connectivity document
    f. Attach Approved Engineering Drawing
    g. Attach other supporting documents
2. Once submitted, this will be pushed to the GNM Engineer who will review the request and the submitted documents. If the documents submission is incorrect or incomplete, the GNM Engineer will reject the request with comments which will notify the REQUESTOR to resubmit the application.
3. The GNM Engineer will be able to edit the details entered by the REQUESTOR. If approved, the GNM Engineer will enter the following details
    a. Edit/fill PMU name
    b. Mnemonic
    c. Substation Type – Dropdown of AIS, GIS and Hybrid
4. The Mnemonic will be accessible to the user by downloading the Mnemonic pdf uploaded by the GNM admin during system setup.
5. Once submitted, the system will assign a new running number for the SLD. The format of the number will be: voltage (first number: 1/2/5) – Substation Type (A/G/H) – Mnemonic – running number. E.g. 1-A-ADAM-001
6. The GNM Engineer will now draw (upload pdf) the new requested SLD and/or Existing remote end PMU/s. This will now be submitted for approval.
7. The new documents will be approved by S/E and DCE in order respectively. They will have the option to Approve or Request Adjustment. If Adjustment is requested, the comments (and additional attachments) will be pushed back to the GNM Engineer to draw and reupload the amended documents.
8. If Approved the new SLD will be pushed back to the REQUESTOR for approval. There will be a new listing screen for the REQUESTOR which will show their Approved SLD.

9. Requestor will now need to check and acknowledge (and upload additional attachments if necessary) these SLD. If they require Adjustment, the workflow will be pushed back to step 5 for the GNM engineer for resubmit the SLD.
10. If Requestor approves the SLD, it will be tied to the station and can now be viewed systemwide.

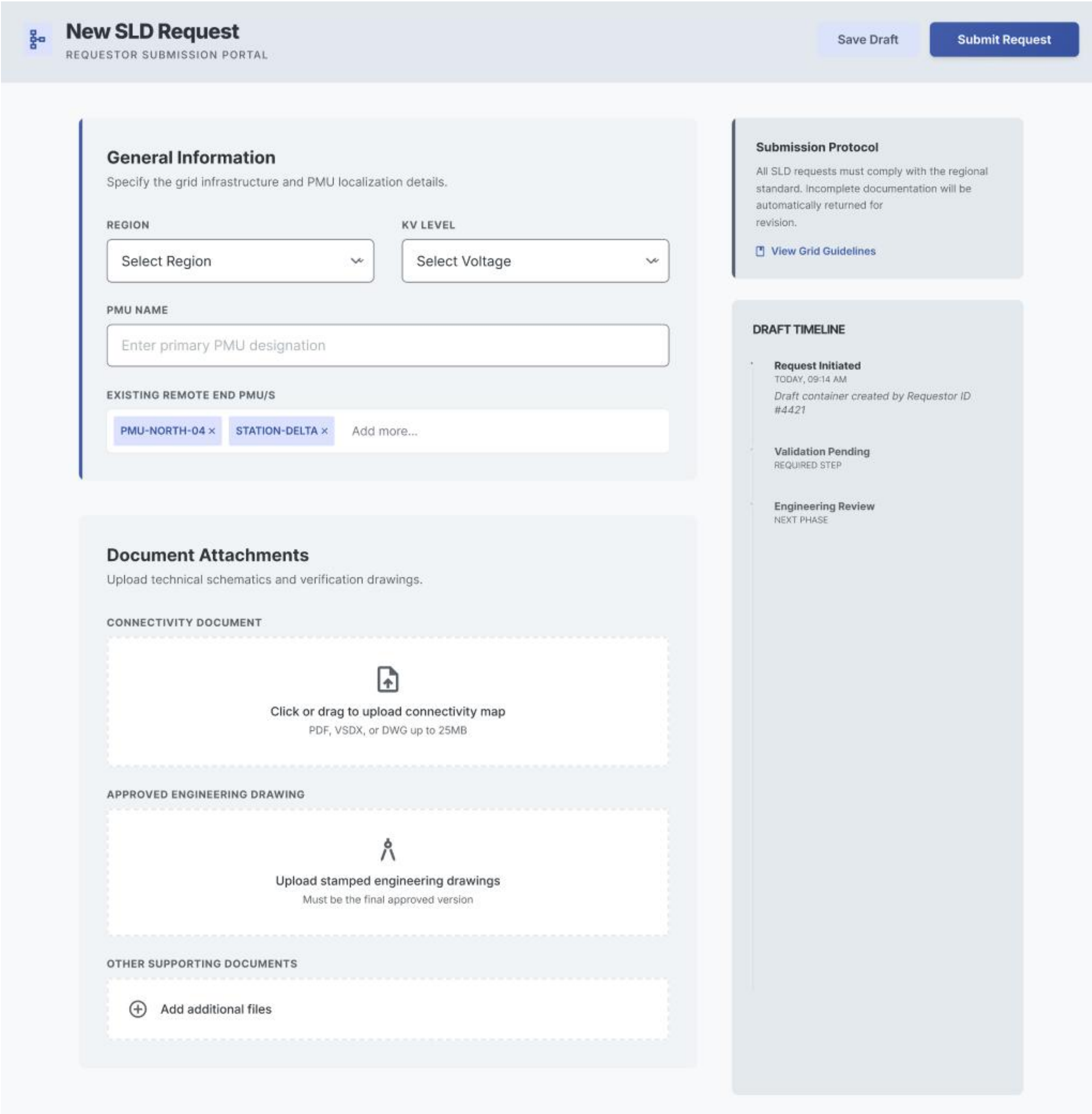


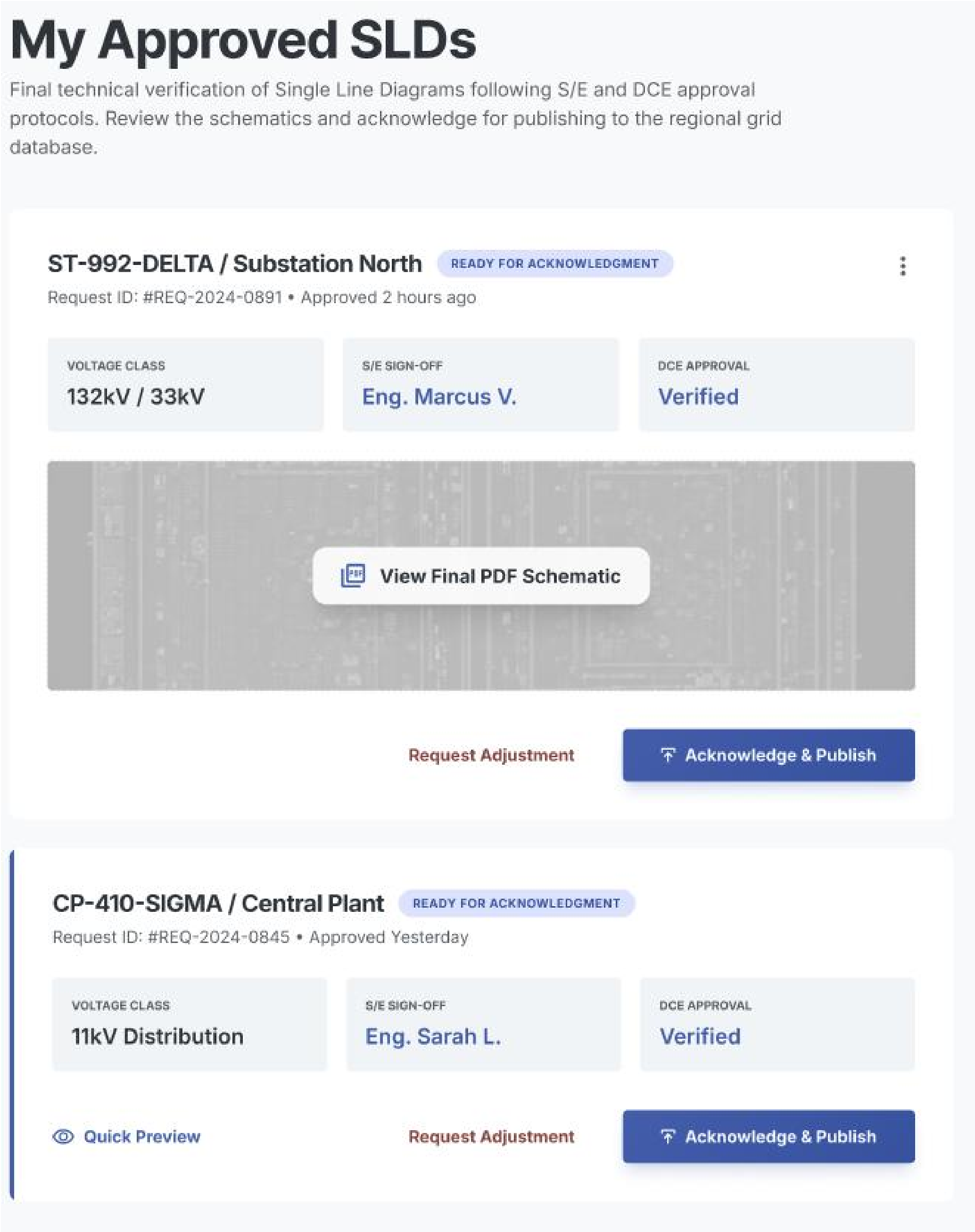
Figure 5.10.1a New SLD Request

5.10.2 Existing SLD

1. The REQUESTOR will select “Existing SLD” and input the following details
    a. Region
    b. Existing PMU – Users will select existing PMU and the Remote end. Multiselect available for multiple lines.
    c. kV level – For every PMU selected
    d. Remark of the changes required – For every PMU selected
    e. Attach Connectivity Document – only once for all the PUMs selected
    f. Attach Approved Engineering Drawing - only once for all the PUMs selected

g. Attach other supporting documents

2. Once submitted, this will be pushed to the GNM Engineer who will review the request and the submitted documents. If the documents submission is incorrect or incomplete, the GNM Engineer will reject the request with comments which will notify the REQUESTOR to resubmit the application.
3. The GNM Engineer will be able to edit the details entered by the REQUESTOR.
4. The GNM Engineer will now draw (upload pdf) the requested existing SLD and Existing remote end PMU/s. This will now be submitted for approval.
5. The new documents will be approved by S/E and DCE in order respectively. They will have the option to Approve or Request Adjustment, with added comments and attachments if necessary. If Adjustment is requested, the comments will be pushed back to the GNM Engineer to draw and reupload the amended documents.
6. If Approved the new SLD will be pushed back to the REQUESTOR for approval.
7. Requestor will now need to check and acknowledge, comment and add attachments as necessary on these SLD. If they require Adjustment, the workflow will be pushed back to step 4 for the GNM engineer for resubmit the SLD.
8. If Requestor approves the SLD, it will be tied to the station and can now be viewed systemwide.



5.10.2a Final Approver by Requestor

5.10.3 Update Module (Post Commission)

1. The GNM ENGINEER will select "Existing SLD" and input the following details
   a. Region
   b. Existing PMU + attach the SLD
   c. Add and select remote end PMU + attach the SLD
   d. Select kV level for each PMU
   e. Remark of the changes
2. The GNM Engineer will now draw (upload pdf) the requested existing SLD and Existing remote end PMU/s. This will now be submitted for approval.
3. The new documents will be approved by S/E and DCE in order respectively. They will have the option to Approve or Request Adjustment. If Adjustment is requested, the comments will be pushed back to the GNM Engineer to draw and reupload the amended documents.

4. If Approved the new SLD will be published in the SLD list for system wide access

5.11 Commissioning Memo

The Commissioning Memo is a new module for ICOMS 2.0, spanning multiple different roles. Commissioning is used to energize new equipment into the electrical grid system. There will be three types of Memos: Commissioning, Emergency Commissioning and Decommissioning.

For the Comm Memo functionality, there will be multiple new roles created, each with a single user per region. These users will be responsible for the approval process of the Comm Memo.

5.11.1 Commissioning Workflow

The Commissioning module allows REQUESTORS to submit a Commissioning Memo (Comm Memo), which is when filled out is passed to multiple users in the GNM division for approval, before being passed onto the GNC user for an execution. The workflow if the Comm Memo creation is as follows, focusing on three parts (Switching program, Data Form and Commissioning Requirement Documents) which can be filled out concurrently.

Switching Program

1. REQUESTOR creates a new Comm Memo from the listing screen. A form needs to be filled showing the Equipment to be commissioned (multi select checkbox), Region – PMU – multiple kV level, working title, proposed date and time, checkbox for New PMU option (if yes, select either TNB PMU, Generation or GCU)
2. REQUESTOR will have the option to select an existing outage to link to this Comm memo. This is the outage that will be used to carry out the Comm Memo and will allow the system to pull the Authorisations number in the outage to be shown after the generation of the Comm Memo. This will be the only field the REQUESTOR can edit after submission.
3. The REQUESTOR will then upload the Switching Program (in pdf format and word format; allowing for multiple file uploads), which will go through a rigorous approval process.
4. The Switching Program will be sent over to the Engineer PIC in the GNM division, who can either reject the entry (which deletes it and ends the workflow), or fill out the Actual Title and edit the data entered by the REQUESTOR before submitting further for approval.
5. The next approvals will be done by the SE, DCE and CE GNM user roles sequentially (GNM roles). They will all be able to edit the data entered at each stage. If at any stage, any one of the users reject the Memo, it will be returned to the Engineer PIC in the Returned Listing screen. The approvers will able to able to upload updated versions of the documents and remove the existing ones.
6. If rejected, the Engineer PIC will revise the Memo based on the rejection comments and return it back to the rejection stage for approval. The Memo will not need to be approved at every stage again; it will return to the user role who rejected it.
7. Once it has passed through the four approvals of the GNM division, the Comm Memo will reach the CE GNC user and DCE user for the next approval step. Both the roles will be able to edit the Comm Memo before pushing for the First two pages of the full memo to be generated.
8. The Comm Memo will be given an auto generated running number, and the front pages will include the following information
    a. Actual Title
    b. Signatures of all signees at the right positions
    c. Current Date

    d. Proposed Time and Date
    e. Requestors Name, designation, and phone number
9. This generation will be viewable in the Interactive document viewer.
10. This data will now be saved in the Engineer PICs Generation tray, where they can view and edit it while waiting for the further information from the REQUESTOR

Data Form – This form needs to be filled out at least 2 weeks before the commissioning date.

11. The REQUESTOR will need to fill out the Data Form for the Comm Memo, following a set template with data entry fields for the customisable sections.
12. REQUESTOR will upload the connectivity document (pdf), which will be sent to the Engineers PICs Generation tray
13. Engineer PIC will vet through the documents and return with comments if there are any errors or wrong information present.
14. Once the form is filled, the system will generate a pdf of this form which will be sent to the Engineers PICs Generation tray. Notification will also be given out to EMS, GSM & GNC.
15. All the info filled in will be placed under a data library.

Commissioning Requirement Documents – Can be uploaded anytime until the last day before Commissioning

16. The Commissioning Requirement Documents will need to be uploaded by the REQUESTOR. There will be set field for the REQUESTOR to upload each document, with reuploads possible for error catching.
    a. IOM Endorsed
    b. Letter from MTEP for Protection Setting
    c. Resident Engineer Certification
    d. Forms G and H respectively
    e. Email chain on metering
    f. Email chain on SCADA
    g. Letter from HGSO (only for Generation type of new PMU)
17. Once Uploaded, they will be pushed to the Comm Memo supporting docs tab, where all GNM users will be able to view for clarity. These documents can be reuploaded to deal with incorrect information.
18. The Engineer PIC can now generate the completed Comm Memo using all the documents and data entered
    a. Memo From Pages
    b. Data From
    c. Connectivity
    d. SW Program
    e. SLD/s from the selected PMU/s (taken from the SLD module)
19. The Generated Comm Memo is pushed to REQUESTOR for viewing, marked as APPROVED. It is no longer editable.
20. The Generated Comm Memo is also pushed the Approved Tray, which is viewable by GNM, GSM, EMS, GNC.
21. GNC, Engineer PIC, SE, DCE, CE GNM, GSM, EMS and Requestor will be notified, they can now download the completed Comm Memo along with the Data Form and the Commissioning Requirement Documents.
22. The GNC is responsible for the final status Change of the Comm Memo shifting it to In progress, On-Soak, Comm Successful or Comm Not Successful.

23. At each stage, all users will be able to view the status of the Comm Memo. The listing screen will allow users to search based on Title and REQUESTOR ID.
24. The Comm Memo generation will be done by previewing the document in an Interactive Document / Smart Document viewer. Users will be able to edit and generate the document at will
25. Users can now shift to the SLD Module and Equipment management to update all information.

The templates and a sample of the Comm Memo have been provided by TNB.

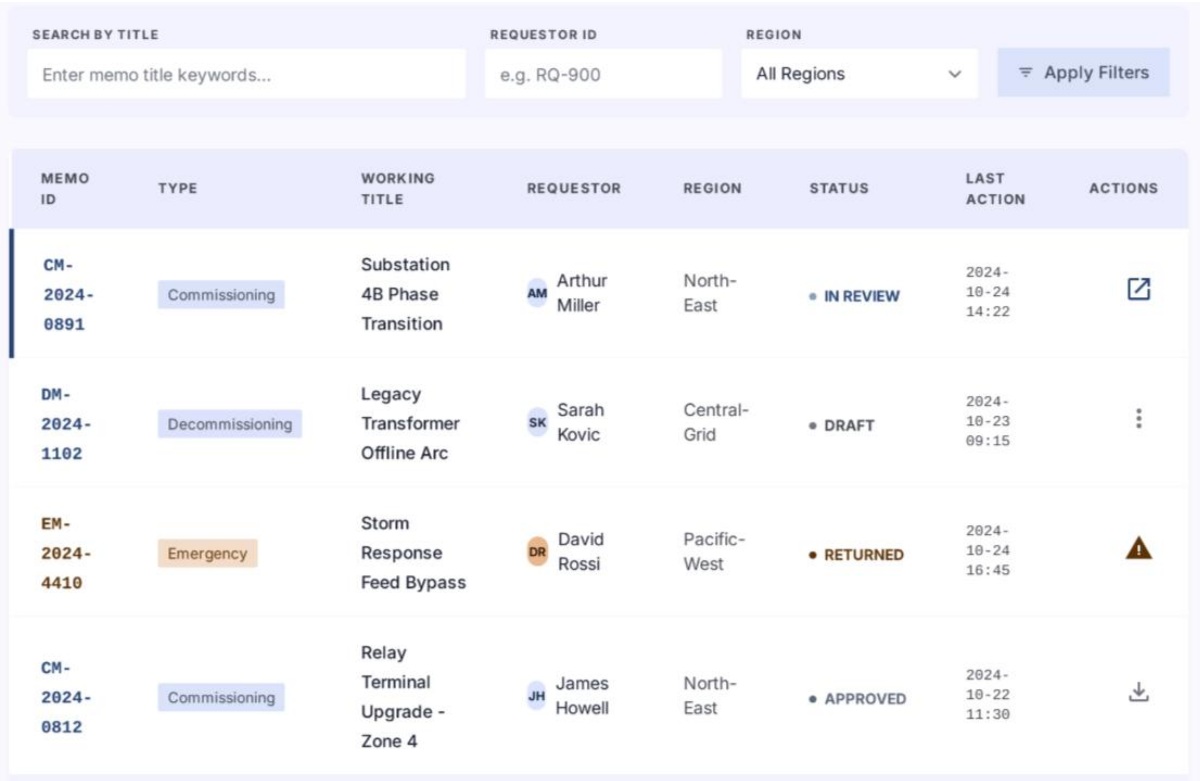


Figure 5.11a Comm Memo listing screen

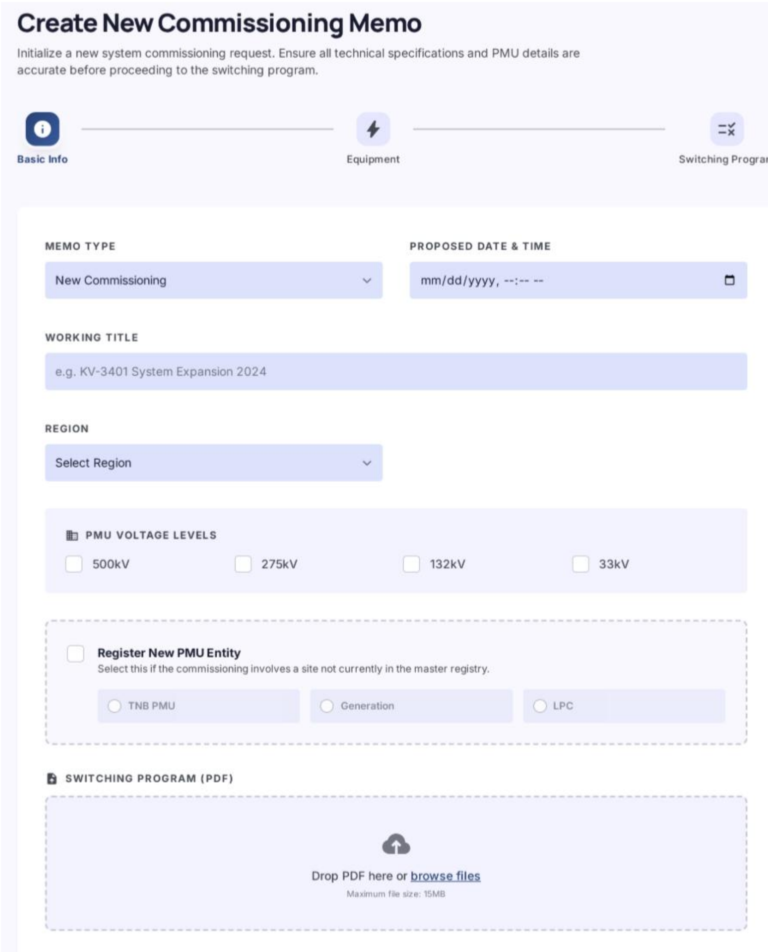


Figure 5.11b Creating New Comm Memo

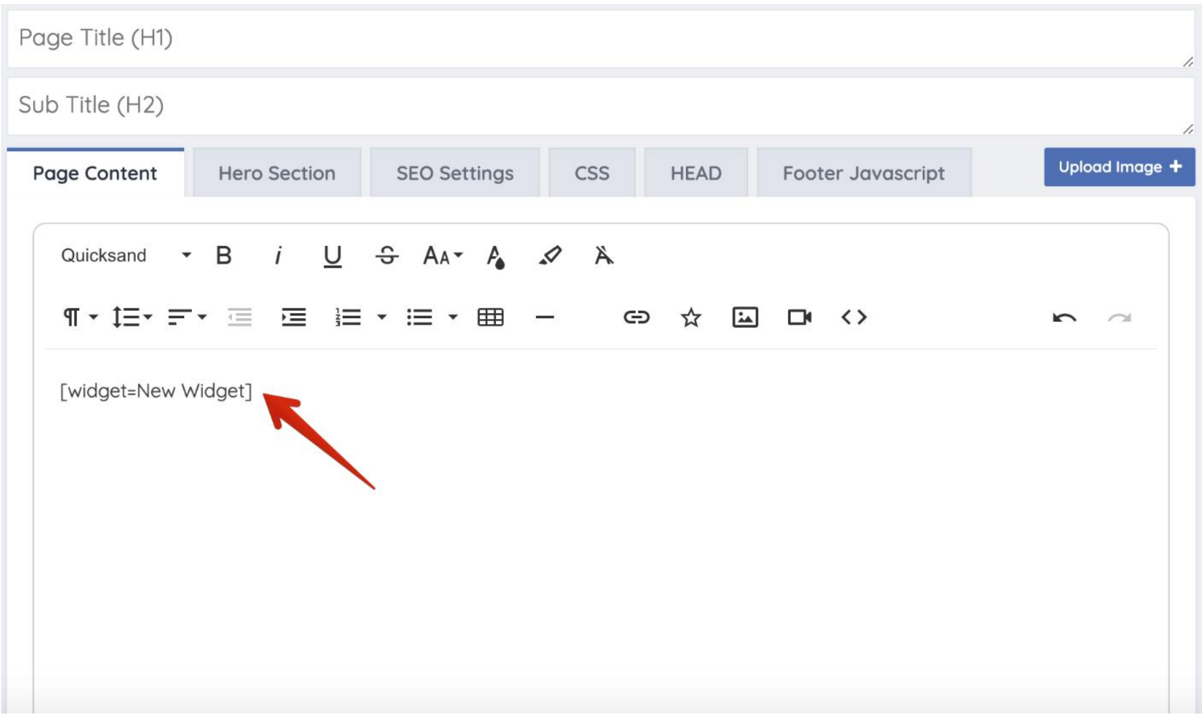


Figure 5.11c Example of Interactive Document Viewer

5.11.1 Emergency Commissioning Workflow

Unlike the standard Commissioning Workflow, Emergency Commissioning workflow only requires the Switching Program Information.

1. REQUESTOR will select Emergency Commissioning from the listing screen and input the following data
    a. Region, PMU and KV level for each Equipment
    b. Equipment/s to be commissioned – a multiselect checkbox
    c. Fill in the title (not actual, can later be edited by the GNC Engineer
    d. Proposed time and date.
2. REQUESTOR will then upload the Switching Program which will be send to the GNC Engineer
3. The GNC Engineer can Return (for edits) or Reject (to delete) the entry.
4. If the GNC Engineer has decided to proceed, they will need to fill out or edit the Full actual tile of the memo, as well as edit or approve all the data filled out by the requestor, as needed.
5. The Comm Memo is then pushed to the CE GNC, who can either Return it to the GNC Engineer or click the "Generate Memo and Send" option which will fill out a running number for the memo and generate the front pages (in the format provided by TNB). The front pages will take the following information from the REQUESTOR data
    a. Title
    b. Signatures
    c. Running Number
    d. Current Date
    e. Proposed Date and Time
    f. Requestors Name, designation and phone number
6. Along with all this the SW Program uploaded and the SLD/s for each PMU/s will be pushed to the final approval tray where it will be viewed by the GNM, GSM, EMS, and GNC Engineer. The GNC Engineer and the Requestor will be notified.
7. The GNC will be responsible to updating the status as work progresses to In progress, On-Soak, Comm Successful or Comm Not Successful.
8. Once the Commissioning has been completed, the NLDC/GNC will upload the Comm Memo in to the listing screen, and select weather the Comm Was Successful or Not Successful.
9. The Emergency Commissioning is now completed and can be viewed by the GNM, SM, EMS and the GNC. Both the GNC and the GNM will be notified.

5.11.1 Decommissioning Workflow

Decommissioning will be done by detaining out the Switching Program

1. REQUESTOR creates a Decommissioning Memo from the listing screen. A form needs to be filled showing the Equipment to be commissioned (multi select checkbox), Region – PMU – multiple kV level, working title, proposed date and time
2. The REQUESTOR will then upload the Switching Program (in pdf format and word format; allowing for multiple file uploads), which will go through a rigorous approval process.
3. The Switching Program will be sent over to the Engineer PIC in the GNM division, who can either reject the entry (which deletes it and ends the workflow), or fill out the Actual Title and edit the data entered by the REQUESTOR before submitting further for approval. There

will be two tick boxes for the Engineer to check for “Changes to SLD” and “Changes to Equipment/Line”

4. The next approvals will be done by the SE, DCE, CE GNM and CE GNC user roles sequentially (GNM roles). They will all be able to edit the data entered at each stage. If at any stage, any one of the users reject the Memo, it will be returned to the Engineer PIC in the Returned Listing screen. The approvers will able to able to upload updated versions of the documents and remove the existing ones.
5. If rejected, the Engineer PIC will revise the Memo based on the rejection comments and return it back to the rejection stage for approval. The Memo will not need to be approved at every stage again; it will return to the user role who rejected it.
6. Once it has passed through the four approvals levels, the CE GNC will be able to push the memo to be generated.
7. The Comm Memo will be given an auto generated running number, and the front pages will include the following information
    a. Actual Title
    b. Signatures of all signees at the right positions
    c. Current Date
    d. Proposed Time and Date
    e. Requestors Name, designation, and phone number
8. This generation will be viewable in the Interactive document viewer.
9. This data will now be saved in the Engineer PICs Generation tray. The Engineer will click the “Generate Memo and send” which will add in the Data Form and Connectivity documents, along with the Switching Program and the SLD/s of the selected PMU/s
10. The DeComm Memo is now generated, and the REQUESTOR will be notified, while the GNM, GSN, EMS and GNC can view it on the listing screen.
11. The GNC will be responsible to updating the status as work progresses to In progress, On-Soak, Comm Successful or Comm Not Successful.
12. Once the Decommissioning has been completed, the NLDC/GNC will upload the Comm Memo in to the listing screen, and select weather the Comm Was Successful or Not Successful.
13. The Decommissioning is now completed and can be viewed by the GNM, SM, EMS and the GNC. Both the GNC and the GNM will be notified.

6.0 USER SIGN OFF – TOMS

User #1 Signature: Name: Designation: Dept.: Date:	User #2 Signature: Name: Designation: Dept.: Date:
User #3 Signature: Name: Designation: Dept.: Date:	User #4 Signature: Name: Designation: Dept.: Date:
User #5 Signature: Name: Designation: Dept.: Date:	User #6 Signature: Name: Designation: Dept.: Date: